$y = \log_b x$, $b > 1$

$Domain = (0, \infty)$ $Range = (-\infty, \infty)$

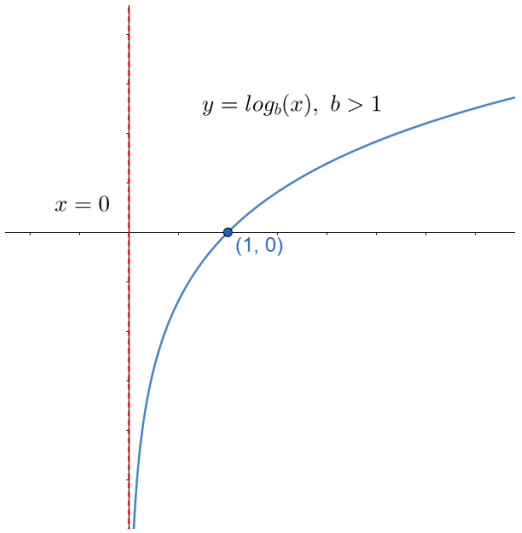
Increasing on $(0, \infty)$

x-intercept: $(1, 0)$

$x \to \infty \Rightarrow y = \log_b x \to \infty$

$x \to 0^+ \Rightarrow y = \log_b x \to -\infty \Rightarrow x = 0$ is a vertical asymptote

Note: $y = \ln x = \log_e x$ where $e \approx 2.7182818$



$y = \log_b x$, $0 < b < 1$

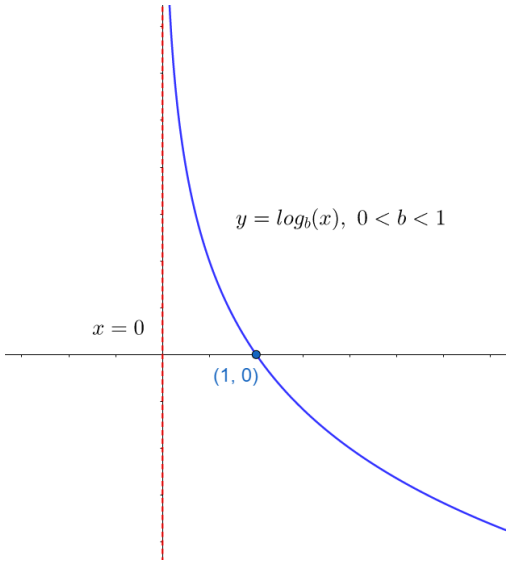
$Domain = (0, \infty)$ $Range = (-\infty, \infty)$

Decreasing on $(0, \infty)$

x-intercept: $(1, 0)$

$x \to \infty \Rightarrow y = \log_b x \to -\infty$

$x \to 0^+ \Rightarrow y = \log_b x \to \infty \Rightarrow x = 0$ is a vertical asymptote



$y = b^x$, $b > 1$

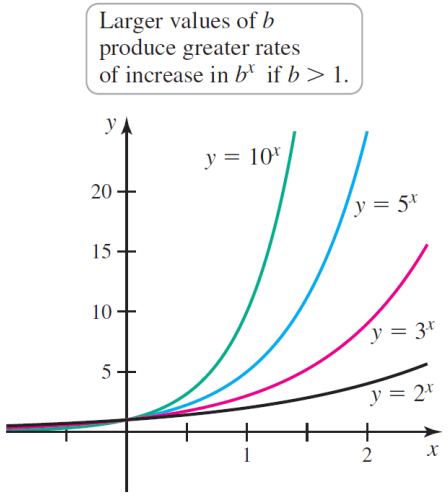
$Domain = (-\infty, \infty)$ $Range = (0, \infty)$

Increasing on $(-\infty, \infty)$

y-intercept: $(0,1)$

$x \to \infty \Rightarrow y = b^x \to \infty$

$x \to -\infty \Rightarrow y = b^x \to 0$ $\Rightarrow y = 0$ is a horizontal asymptote



$y = b^x$, $0 < b < 1$

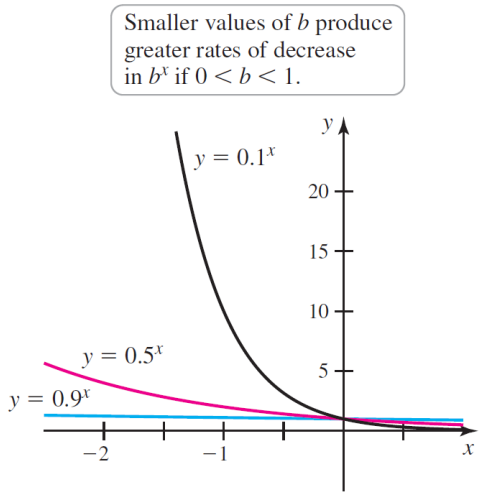
$Domain = (-\infty, \infty)$ $Range = (0, \infty)$

Decreasing on $(-\infty, \infty)$

y-intercept: $(0,1)$

$x \to \infty \Rightarrow y = b^x \to 0$ $\Rightarrow y = 0$ is a horizontal asymptote

$x \to -\infty \Rightarrow y = b^x \to \infty$



The **Floor Function** $f(x) = \lfloor x \rfloor$ is a function which gives the greatest integer less than or equal to x.

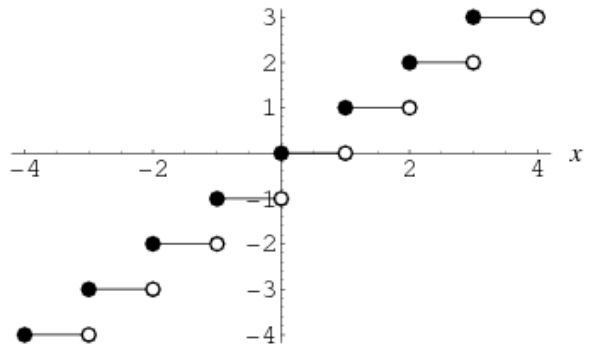
$Domain = (-\infty, \infty)$

$Range = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\} = \mathbb{Z}$

Increasing on $(-\infty, \infty)$

$x \to \infty \Rightarrow f(x) = \lfloor x \rfloor \to \infty$ $\quad$ $x \to -\infty \Rightarrow f(x) = \lfloor x \rfloor \to -\infty$

$f(x) = \lfloor x \rfloor$



The **Ceiling Function** $f(x) = \lceil x \rceil$ is a function which gives the smallest integer greater than or equal to x.

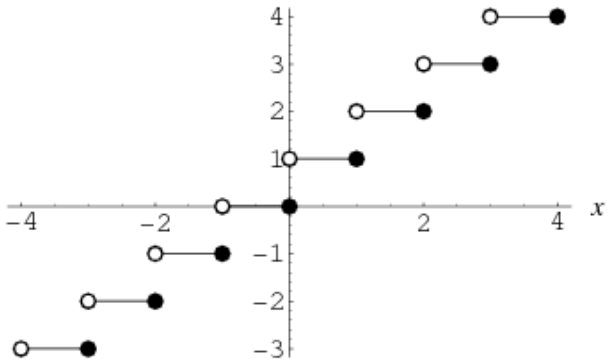
$Domain = (-\infty, \infty)$

$Range = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\} = \mathbb{Z}$

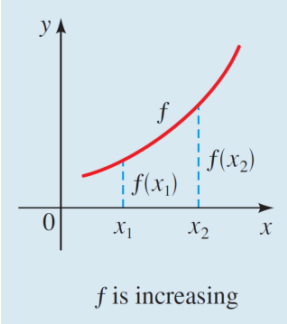
Increasing on $(-\infty, \infty)$

$x \to \infty \Rightarrow f(x) = \lceil x \rceil \to \infty$ $\quad$ $x \to -\infty \Rightarrow f(x) = \lceil x \rceil \to -\infty$

$f(x) = \lceil x \rceil$



Definition: A function is called <u>increasing</u> on an interval if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ for any x_1 and x_2 in the interval.



Can you think of a function that is increasing on its domain?

The following functions are increasing on their domains:

$$y = \log_b x \ , \ b > 1$$

$$y = b^x \ , \ b > 1$$

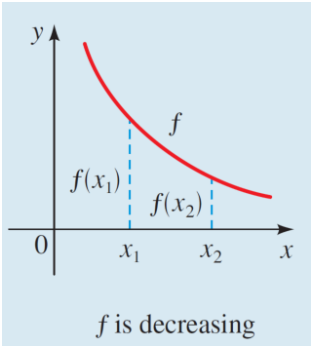
$y = x^n$, where n is odd: $y = x$, $y = x^3$, . . .

$$y = \sqrt[n]{x} \ : \ y = \sqrt{x} \ , y = \sqrt[3]{x} \ , \ldots$$

$$y = \sin^{-1} x$$

$$y = \tan^{-1} x$$

Definition: A function is called <u>decreasing</u> on an interval if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$ for any x_1 and x_2 in the interval.



Can you think of a function that is decreasing on its domain?

The following functions are decreasing on their domains:

$$y = \log_b x \ , \ 0 < b < 1$$

$$y = b^x \ , \ 0 < b < 1$$

$$y = \cos^{-1} x$$

$$y = \cot^{-1} x$$

The **absolute value function**: $f(x)=|x|$

$Domain=(-\infty,\infty)$ $Range=[0,\infty)$

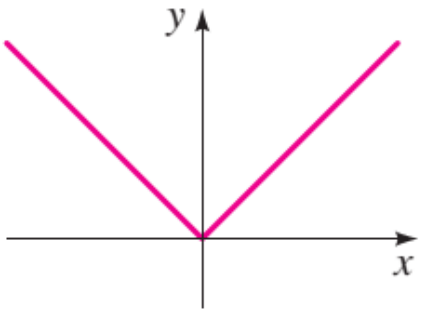
Increasing on $(0,\infty)$

Decreasing on $(-\infty,0)$

x and y-intercept: $(0,0)$

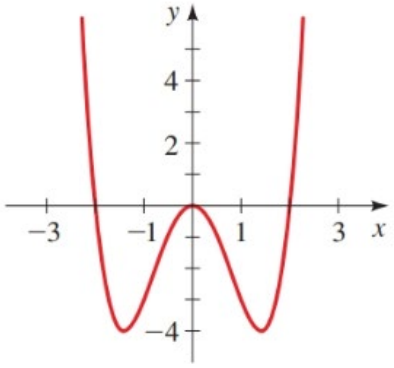
$x\to\infty \Rightarrow f(x)=|x|\to\infty$

$x\to-\infty \Rightarrow f(x)=|x|\to\infty$



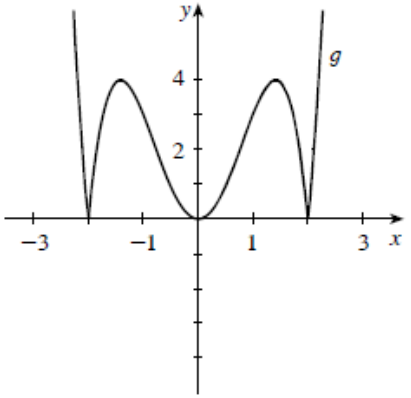
$f(x)=|x|$

You are given the graph of a function $y=f(x)$.

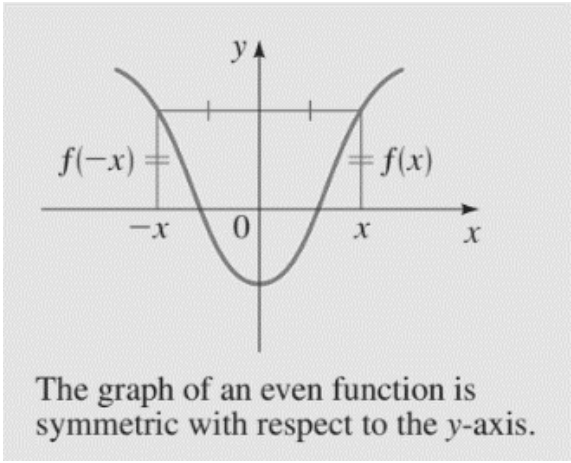


What would the graph of $g(x)=|f(x)|$ look like?

Graph of $g(x)=|x|$.



f is <u>even</u> if $f(-x) = f(x)$ for all x in the domain of f.



The graph of an even function is symmetric with respect to the y-axis.

Can you think of examples of even functions?

The following functions are even:

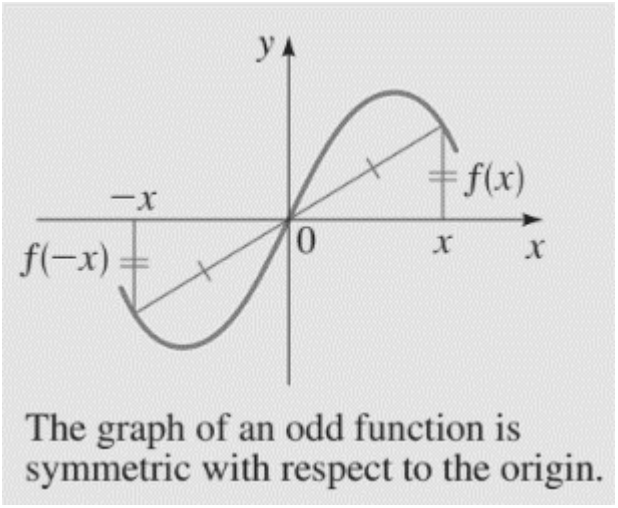
$$y = x^2$$

$$y = \frac{1}{x^2}$$

$$y = \cos x$$

$$y = \sec x$$

f is <u>odd</u> if $f(-x) = -f(x)$ for all x in the domain of f.



The graph of an odd function is symmetric with respect to the origin.

Can you think of examples of odd functions?

The following functions are odd:

$$y = x^3$$

$$y = \frac{1}{x}$$

$$y = \sin x$$

$$y = \csc x$$

$$y = \tan x$$

$$y = \cot x$$

Rational Function: $y = \frac{1}{x}$

$Domain = (-\infty, 0) \cup (0, \infty)$ $\quad$ $Range = (-\infty, 0) \cup (0, \infty)$

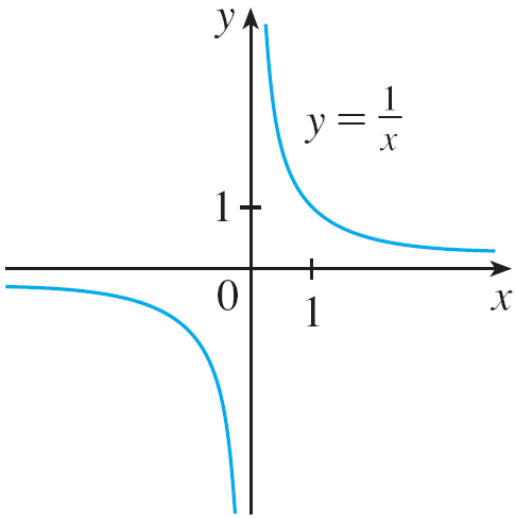
Decreasing on $(-\infty, 0) \cup (0, \infty)$

$x \to \infty \Rightarrow y = \frac{1}{x} \to 0$, $x \to -\infty \Rightarrow y = \frac{1}{x} \to 0$

$\Rightarrow$ $y = 0$ is a horizontal asymptote

$x \to 0^+ \Rightarrow y = \frac{1}{x} \to \infty$, $x \to 0^- \Rightarrow y = \frac{1}{x} \to -\infty$

$\Rightarrow x = 0$ is a vertical asymptote



Odd Function. Symmetric with respect to the origin.

Rational Function: $y = \frac{1}{x^2}$

$Domain = (-\infty, 0) \cup (0, \infty)$ $\quad$ $Range = (0, \infty)$

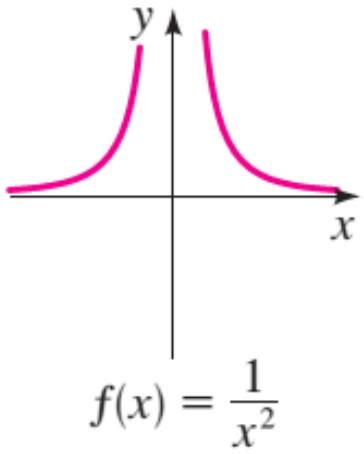
Increasing on $(-\infty, 0)$ $\quad$ Decreasing on $(0, \infty)$

$x \to \infty \Rightarrow y = \frac{1}{x^2} \to 0$, $x \to -\infty \Rightarrow y = \frac{1}{x^2} \to 0$

$\Rightarrow y = 0$ is a horizontal asymptote

$x \to 0^+ \Rightarrow y = \frac{1}{x^2} \to \infty$, $x \to 0^- \Rightarrow y = \frac{1}{x^2} \to \infty$

$\Rightarrow x = 0$ is a vertical asymptote



Even function. Symmetric with respect to the y-axis.

Power Functions: $y = x^n$, n is even

$Domain = (-\infty, \infty)$ $Range = [0, \infty)$

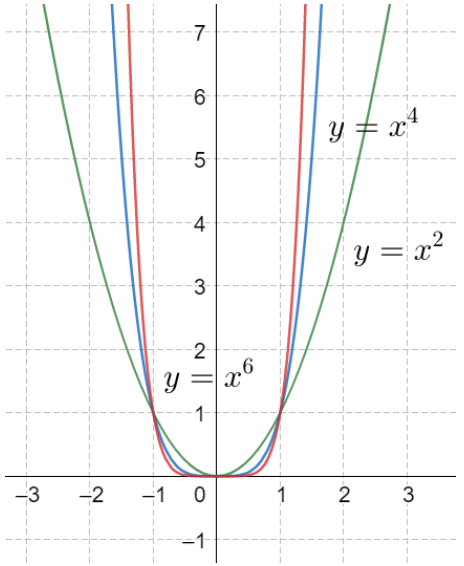
Increasing on $(0, \infty)$ Decreasing on $(-\infty, 0)$

x and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = x^n \to \infty$ $x \to -\infty \Rightarrow y = x^n \to \infty$

Even Function.

Symmetric with respect to the *y*-axis.



Power Function: $y = x^n$, n is odd

$Domain = (-\infty, \infty)$ $Range = (-\infty, \infty)$

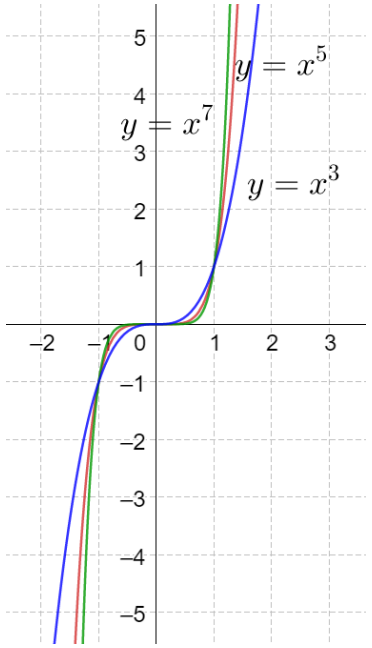
Increasing on $(-\infty, \infty)$

x and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = x^n \to \infty$ $x \to -\infty \Rightarrow y = x^n \to -\infty$

Odd function.

Symmetric with respect to the origin.



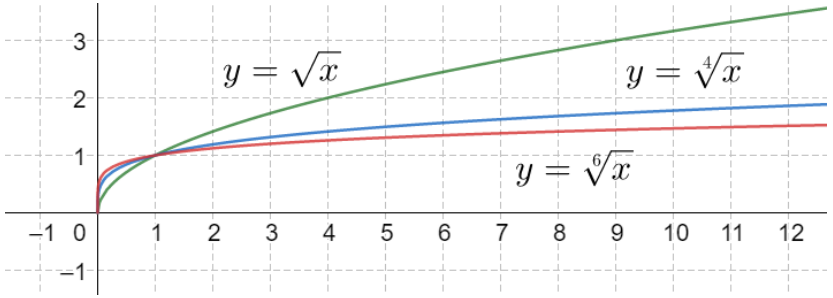
Radical Function: $y = \sqrt[n]{x} = x^{1/n}$, n is even

$Domain = [0, \infty)$ $Range = [0, \infty)$

Increasing on $(0, \infty)$

x and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = \sqrt[n]{x} \to \infty$



Radical Function: $y = \sqrt[n]{x} = x^{1/n}$, n is odd

$Domain = (-\infty, \infty)$ $Range = (-\infty, \infty)$

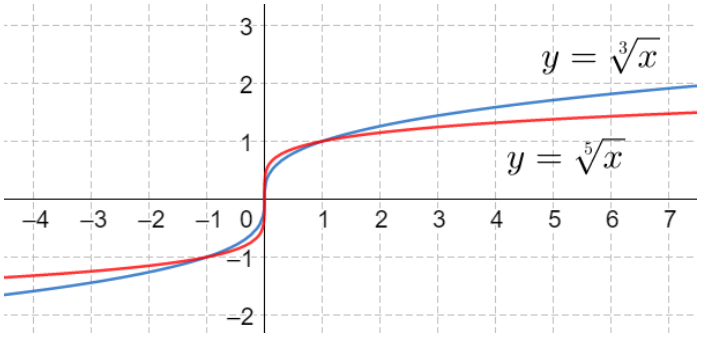
Increasing on $(-\infty, \infty)$

x and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = \sqrt[n]{x} \to \infty$ $x \to -\infty \Rightarrow y = \sqrt[n]{x} \to -\infty$

Odd function.

Symmetric with respect to the origin.



Linear Function: $y = mx + b,\ m > 0$

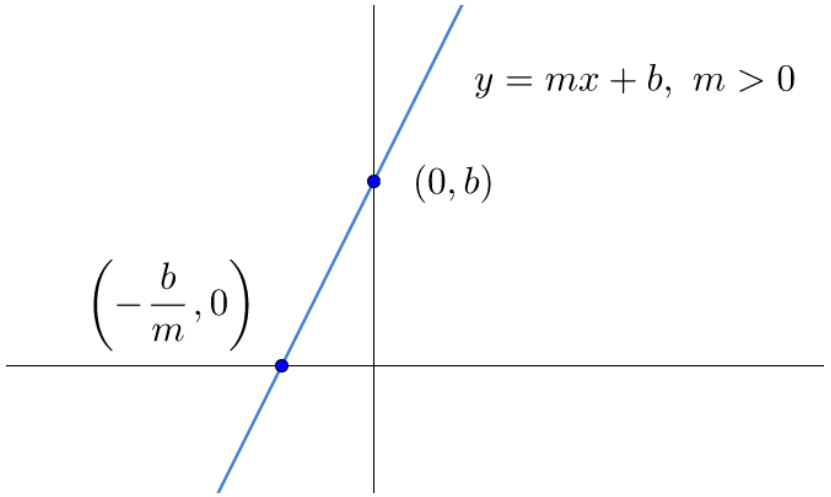
$Domain = (-\infty, \infty)$ $\quad$ $Range = (-\infty, \infty)$

If $m > 0$, then $y = mx + b$ is increasing on $(-\infty, \infty)$

x-intercept: $\left(-\frac{b}{m}, 0\right)$ $\quad$ y-intercept: $(0, b)$

$x \to \infty \Rightarrow y = mx + b \to \infty$

$x \to -\infty \Rightarrow y = mx + b \to -\infty$



Linear Function: $y = mx + b,\ m < 0$

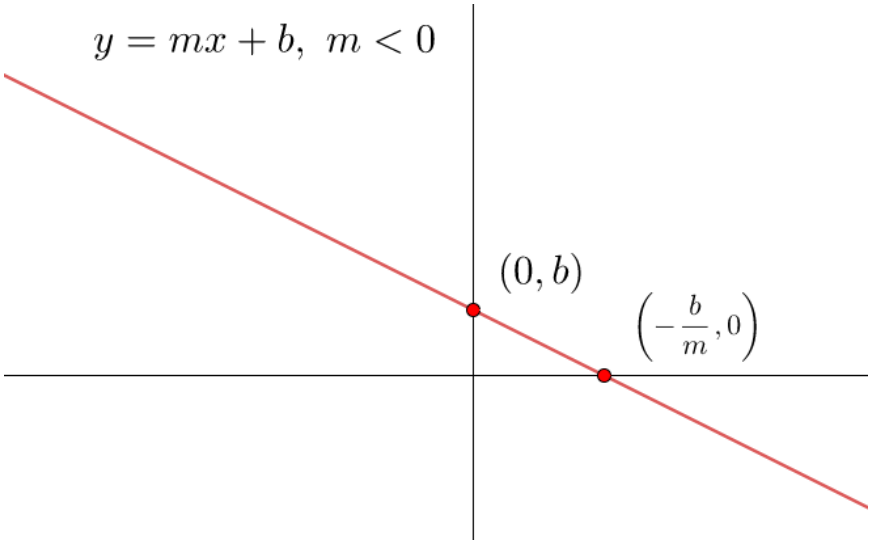
$Domain = (-\infty, \infty)$ $\quad$ $Range = (-\infty, \infty)$

If $m < 0$, then $y = mx + b$ is decreasing on $(-\infty, \infty)$

x-intercept: $\left(-\frac{b}{m}, 0\right)$ $\quad$ y-intercept: $(0, b)$

$x \to \infty \Rightarrow y = mx + b \to -\infty$

$x \to -\infty \Rightarrow y = mx + b \to \infty$

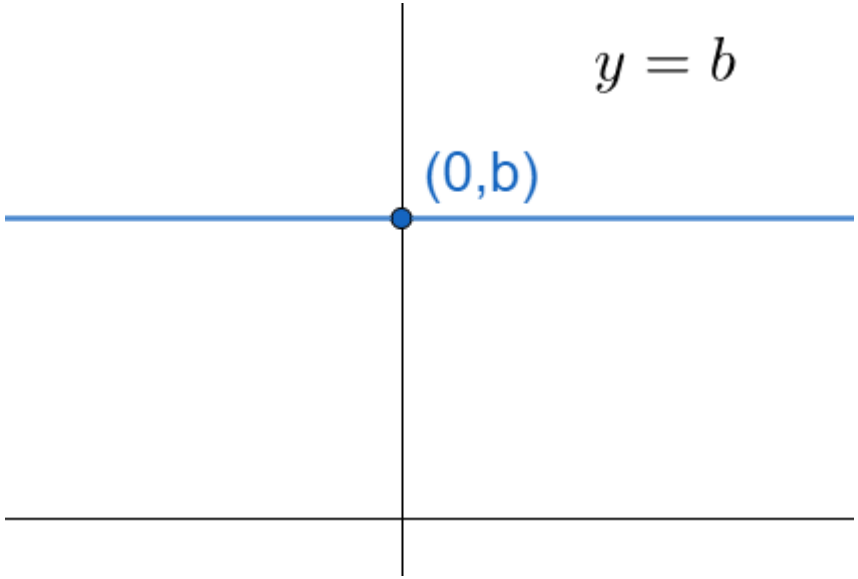


$y = b$

$Domain = (-\infty, \infty)$ $Range = \{b\}$

$y = b$ is constant on $(-\infty, \infty)$

y-intercept: $(0, b)$

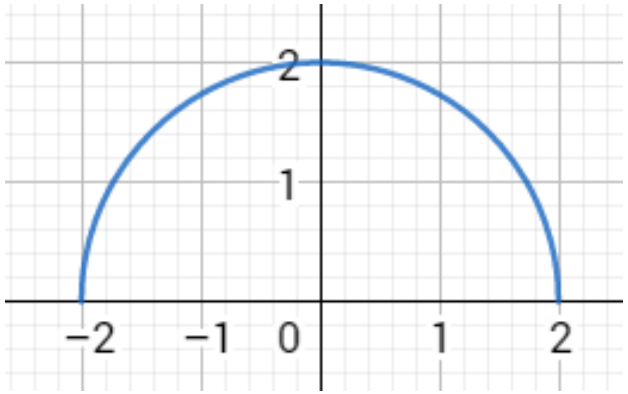


Semi-Circle (Upper) centered at Origin: $y = \sqrt{r^2 - x^2}$

Example: $y = \sqrt{4 - x^2}$

Domain: $[-2, 2]$

Range $[0, 2]$

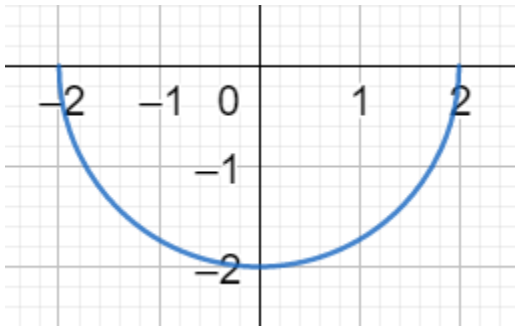


Semi-Circle(Lower) centered at Origin: $y = -\sqrt{r^2 - x^2}$

Example: $y = -\sqrt{4 - x^2}$

Domain: $[-2, 2]$

Range $[-2, 0]$



$y = \sin(x)$

$Domain = (-\infty, \infty)$ $Range = [-1,1]$

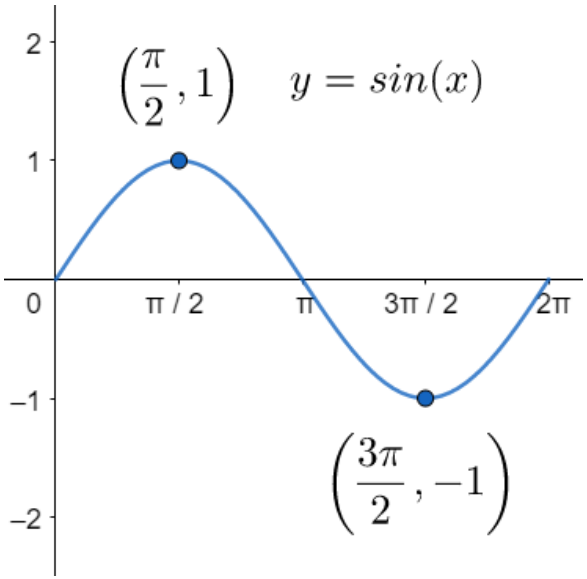
$Period = 2\pi$

x-intercepts: $(\pi k, 0)$ where k is an integer

y-intercept: $(0,0)$

Odd function.

Symmetric with respect to the origin.



$y = \cos(x)$

$Domain = (-\infty, \infty)$ $Range = [-1,1]$

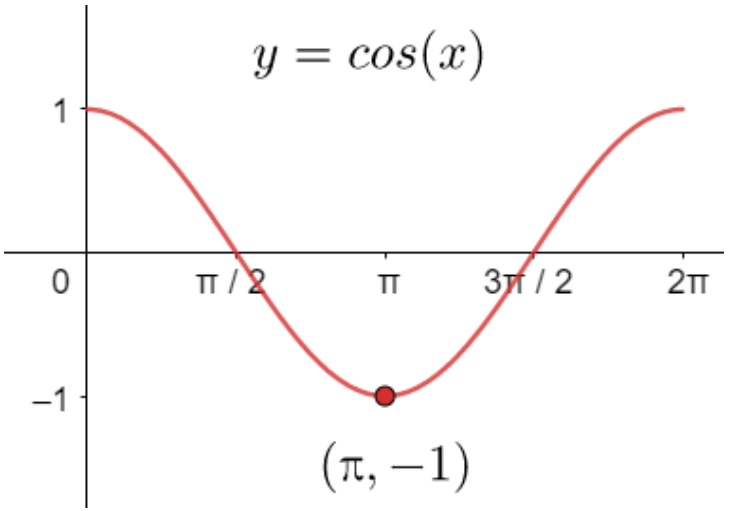
$Period = 2\pi$

x-intercepts: $\left(\frac{\pi}{2}(2k+1), 0\right)$ where k is an integer

y-intercept: $(0,1)$

Even function.

Symmetric with respect to the y-axis.



$y = \csc(x)$

$Domain = \{x \in \mathbb{R} | x \neq \pi k \text{ where } k \text{ is an integer}\}$

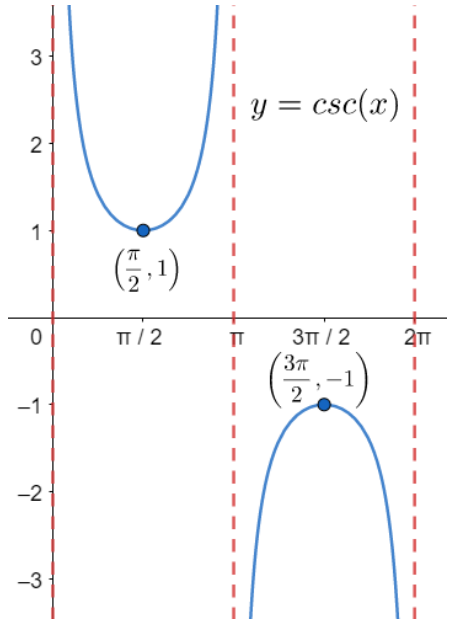
$Range = (-\infty, -1] \cup [1, \infty)$

$Period = 2\pi$

vertical asymptotes: $x = \pi k$ where k is an integer

Odd function.

Symmetric with respect to the origin.



$y = \sec(x)$

$Domain = \left\{x \in \mathbb{R} | x \neq \frac{\pi}{2}(2k+1) \text{ where } k \text{ is an integer}\right\}$

$Range = (-\infty, -1] \cup [1, \infty)$

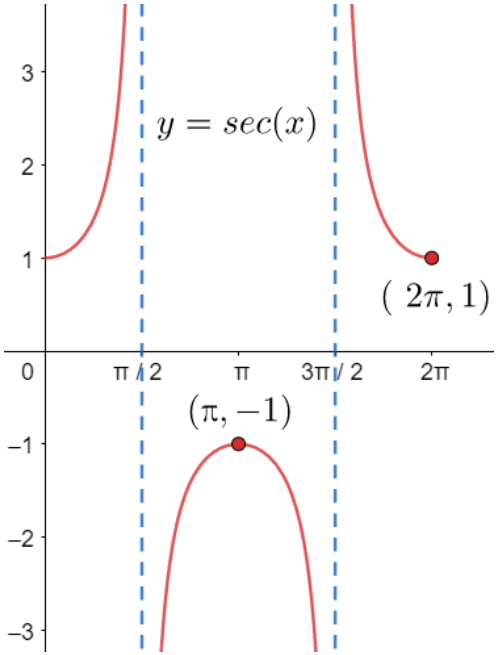
$Period = 2\pi$

y-intercept: $(0,1)$

vertical asymptotes: $x = \frac{\pi}{2}(2k+1)$ where k is an integer

Even function.

Symmetric with respect to the y-axis.



$y = \tan(x)$

$Domain = \left\{ x \in \mathbb{R} \middle| x \neq \frac{\pi}{2}(2k+1) \text{ where } k \text{ is an integer} \right\}$

$Range = (-\infty, \infty)$

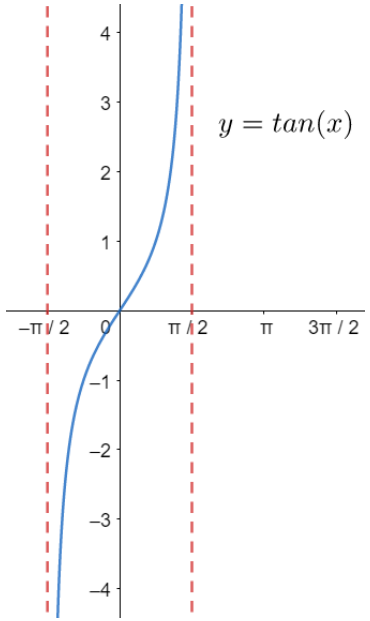
$Period = \pi$

vertical asymptotes: $x = \frac{\pi}{2}(2k+1)$ where k is an integer

x-intercepts: $(\pi k, 0)$ where k is an integer

y-intercept: $(0,0)$

Odd function. Symmetric with respect to the origin.



$y = \cot(x)$

$Domain = \left\{ x \in \mathbb{R} \middle| x \neq \pi k \text{ where } k \text{ is an integer} \right\}$

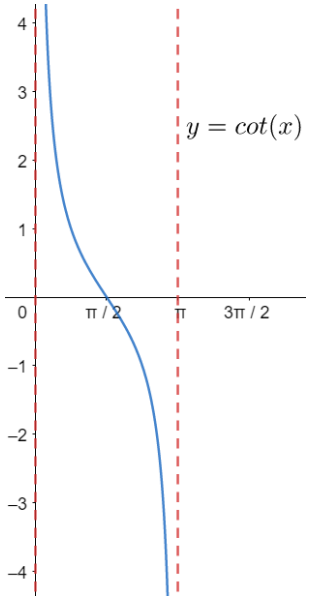
$Range = (-\infty, \infty)$

$Period = \pi$

vertical asymptotes: $x = \pi k$ where k is an integer

x-intercepts: $\left(\frac{\pi}{2}(2k+1), 0 \right)$ where k is an integer

Odd function. Symmetric with respect to the origin.

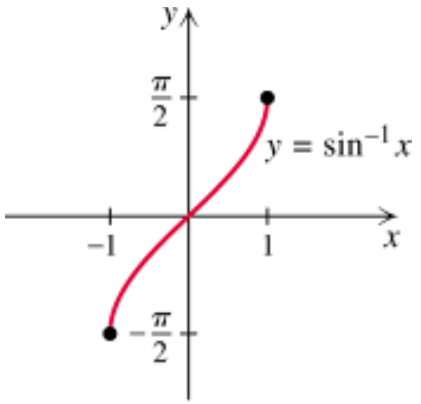


$y = \sin^{-1} x$

$Domain = [-1,1]$ $Range = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

x-intercepts: $(0,0)$

y-intercept: $(0,0)$



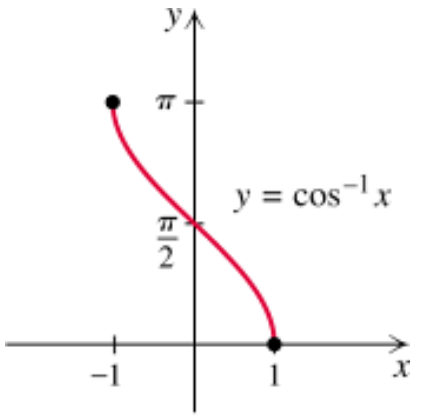
Domain [−1, 1]
Range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$y = \cos^{-1} x$

$Domain = [-1,1]$ $Range = [0,\pi]$

x-intercepts: $(1,0)$

y-intercept: $\left(0, \frac{\pi}{2}\right)$



Domain [−1, 1]
Range [0, π]

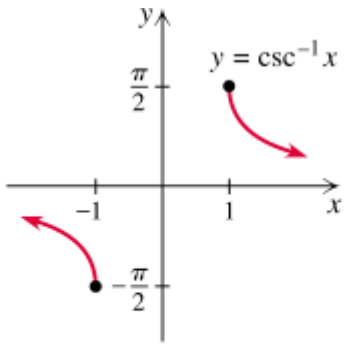
$y = \csc^{-1} x$

$Domain = (-\infty, -1] \cup [1, \infty)$ $\quad$ $Range = \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

$x \to \infty \Rightarrow y = \csc^{-1} x \to 0$

$x \to -\infty \Rightarrow y = \csc^{-1} x \to 0$

$y = 0$ is a horizontal asymptote



Domain $(-\infty, -1] \cup [1, \infty)$
Range $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

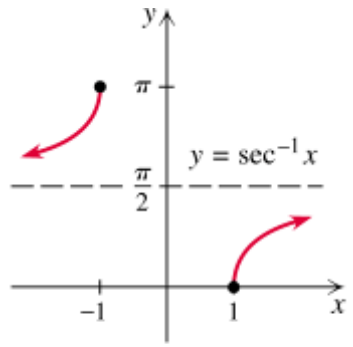
$y = \sec^{-1} x$

$Domain = (-\infty, -1] \cup [1, \infty)$ $\quad$ $Range = \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

x-intercepts: $(1, 0)$

$x \to \infty \Rightarrow y = \sec^{-1} x \to \frac{\pi}{2}$ $\quad$ $x \to -\infty \Rightarrow y = \sec^{-1} x \to \frac{\pi}{2}$

$y = \frac{\pi}{2}$ is a horizontal asymptote



Domain $(-\infty, -1] \cup [1, \infty)$
Range $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

$y = \tan^{-1} x$

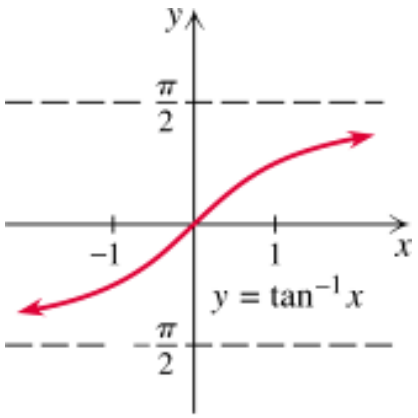
$Domain = (-\infty, \infty)$

$Range = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

x-and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = \tan^{-1} x \to \frac{\pi}{2} \quad \Rightarrow \quad y = \frac{\pi}{2}$ is a horizontal asymptote

$x \to -\infty \Rightarrow y = \tan^{-1} x \to -\frac{\pi}{2} \quad \Rightarrow \quad y = -\frac{\pi}{2}$ is a horizontal asymptote



Domain $(-\infty, \infty)$

Range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$y = \cot^{-1} x$

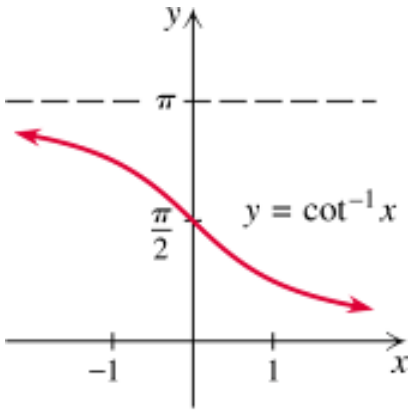
$Domain = (-\infty, \infty)$

$Range = (0, \pi)$

y-intercept: $\left(0, \frac{\pi}{2}\right)$

$x \to \infty \Rightarrow y = \cot^{-1} x \to 0 \quad \Rightarrow \quad y = 0$ is a horizontal asymptote

$x \to -\infty \Rightarrow y = \cot^{-1} x \to \pi \Rightarrow \quad y = \pi$ is a horizontal asymptote



Domain $(-\infty, \infty)$

Range $(0, \pi)$